

FIPS 140-3 Implementation Guidance Checklist

Module: pqc_crypto_module v0.1.0 **Security Level Target:** 1 (Software) **Date:** 2026-04-28

Disclaimer: This package does not imply FIPS 140-3 validation or CMVP certification.

Assessment Key

Status	Meaning
PASS	Requirement met with evidence
PARTIAL	Partially implemented, gaps identified
FAIL	Not implemented or insufficient
N/A	Not applicable to this module type/level

Checklist

#	IG Area	Status	Evidence	Gap	Owner	Next A
1	Module Specification	PARTIAL	Module name, version, algorithm list defined in Cargo.toml and lib.rs. Boundary enforced by tests/crypto_boundary.rs.	No formal Module Specification document (Security Policy). Missing: ports/interfaces table, service descriptions, mode of operation narrative. See F-008, F-010.	Module owner	Author formal Module Specification document FIPS 1 Section
2		PASS				

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	Cryptographic Module Boundary		11 source files in crates/pqc_crypto_module/src/.tests/crypto_boundary.rs enforces 100% compliance (189/189 files). All crypto operations route through the module's public API.	None for the code. Boundary definition needs to appear in the formal spec document (see #1).	Module owner	Include bounda diagram Module Specifici
3	Approved Cryptographic Services	PARTIAL	ML-DSA-65 (FIPS 204): fully functional, sign/verify integrated end-to-end. SHA3-256 (FIPS 202): functional, used in self-tests and hashing. ML-KEM-768 (FIPS 203): placeholder only.	ML-KEM-768 is listed as approved but not functional. See F-001.	Module owner	Either comple ML-KI remove approv
4	Non-Approved Cryptographic Services	PARTIAL	Ed25519, SHA-256, HMAC-SHA256 identified as non-approved. Runtime-gated via SIGNING_ALGORITHM env var. Feature-flag exclusion documented.	No FSM-enforced blocking of non-approved algorithms in Approved mode. See F-006.	Module owner	Add ex FSM st check a entry p of all n approv code p
5	Roles and Authentication	PARTIAL	Implicit single role (Crypto Officer = process owner). No separate User role defined.	FIPS 140-3 requires at least Crypto Officer and User roles. No role-based authentication within the module (acceptable at Level 1 but must be documented).	Module owner	Docum roles in Securit Policy. process initializ module = any c after in
6	Finite State Machine (FSM)	PASS	4 states: Uninitialized, SelfTesting, Approved, Error. AtomicU8 with SeqCst ordering. Transitions well-defined. Error is terminal.	No formal state diagram in documentation. No exhaustive invalid-transition tests. See F-007.	Module owner	Add sta diagram Securit Policy. exhaus transiti tests.
7	Self-Tests	PARTIAL	Power-on self-tests: SHA3 KAT, ML-DSA sign/verify/corrupt KAT, ML-KEM encaps/decaps KAT, continuous RNG test. Self-	No ACVP vectors. Self-test vectors are internal. See F-002. No	Module owner	Integra ACVP vectors condition

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			tests must pass before FSM enters Approved.	conditional self-tests on algorithm parameter changes.		self-test hooks.
8	Error Handling	PASS	Self-test failure transitions FSM to Error. All crypto operations check FSM state and return CryptoError on failure. Fail-closed behavior confirmed.	Recovery path undocumented. See F-009.	Module owner	Document Error and termination in Security Policy.
9	Key Management	PARTIAL	MldsPrivateKey (4032 bytes) with ZeroizeOnDrop. MlKemPrivateKey with ZeroizeOnDrop. MlKemSharedSecret with ZeroizeOnDrop. Keys generated via OsRng.	No key import/export format specification. No key usage period enforcement. No mlock. See F-004.	Module owner	Document key lifecycle. Add mlock. Define transport needed.
10	Zeroization	PARTIAL	ZeroizeOnDrop on all private key types and shared secrets. zeroize crate uses write_volatile + fence.	No mlock (key pages could swap). Compiler optimization concern is theoretical. See F-004, F-005.	Module owner	Add mlock. Add assembly verification. CI step.
11	Entropy / RNG	PARTIAL	OsRng via getrandom crate. Continuous RNG test (generate, check non-zero/non-repeating). CryptoError::RngFailure on failure. No fallback RNG.	No SP 800-90A DRBG within module boundary. No SP 800-90B entropy source documentation. See F-003.	Module owner	Add H. DRBG documentation. RNG justification per SP 800-90.
12	Operational Environment	PARTIAL	Targets general-purpose OS (Linux, macOS). Rust nightly toolchain. Single-process, multi-threaded.	No documented minimum OS requirements. No attestation of OS integrity (acceptable at Level 1). Need to specify supported OS versions.	Module owner	Document support list and toolchain requirements.
13	Physical Security	N/A	Software-only module at Security Level 1. No physical enclosure.	N/A	N/A	N/A

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14	Non-Invasive Attack Mitigation	N/A	Not required at Security Level 1.	N/A	N/A	N/A
15	Lifecycle Assurance	PARTIAL	Version control (git). CI/CD with cargo fmt, cargo clippy, cargo test. Cargo.lock committed.	Reproducible build not yet proven. See F-011. No formal CM plan document.	Module owner	Execute clean-r build. Document CM proced
16	Design Assurance	PARTIAL	Source code available. Test suite with KATs. Boundary test. Architecture documented in CLAUDE.md.	No formal design document per FIPS 140-3 requirements. No formal code review records (lab expects documented evidence).	Module owner	Produce design document. Archive review records
17	Guidance Documents	FAIL	No Security Policy. No Crypto Officer Guide. No User Guide.	All three are mandatory deliverables for CMVP submission. See F-010.	Module owner	Author Security Policy (combined CO + U guide) 800-14 series.

Summary

Status	Count
PASS	3
PARTIAL	12
FAIL	1
N/A	2

Key blockers for lab intake: 1. Guidance Documents (FAIL) — Security Policy is mandatory 2. Approved Services — ML-KEM placeholder must be resolved 3. Self-Tests — ACVP vectors required 4. RNG — SP 800-90A/B compliance gap

End of FIPS 140-3 IG checklist.